

## BELFORT FONTAINE, France

WMO: 072950

Lat: 47.633N

Lon: 6.867E

Elev: 423

StdP: 96.35

Time Zone: 1.00 (EUC)

Period: 99-09

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.8       | -6.0 | -10.5                         | 1.6 | -6.2 | -8.9 | 1.9 | -4.7 | 9.4                   | 8.9  | 8.4 | 8.2  | 1.2                   | 80   | 0.478 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.1                   | 30.1            | 19.8 | 28.2 | 19.3 | 26.6 | 18.9 | 21.0                | 27.9 | 20.2 | 26.8 | 19.4 | 25.4 | 1.8                  | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.6                            | 14.1 | 23.9 | 17.8 | 13.5 | 22.9 | 17.1 | 12.9 | 22.0 | 62.5          | 28.2 | 59.7 | 26.9 | 56.9 | 25.6 | 23.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.0               | 5.9  | 5.0 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |     | -11.0                      | 22.4 | 1.9                | 1.0 | -12.4                                              | 23.1 | -13.6      | 23.7 | -14.6      | 24.3 | -16.1      | 25.0 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.4        | 1.5   | 2.8   | 6.2   | 10.0  | 14.5  | 17.2  | 18.9  | 18.7  | 15.4  | 11.1  | 5.5   | 2.0   | (6)  |
|                                                                        | DBStd       | 7.36        | 4.23  | 4.04  | 4.13  | 4.01  | 3.71  | 3.67  | 3.37  | 3.39  | 3.40  | 3.68  | 3.63  | 3.92  | (7)  |
|                                                                        | HDD10.0     | 1075        | 264   | 202   | 131   | 49    | 6     | 1     | 0     | 0     | 2     | 33    | 138   | 248   | (8)  |
|                                                                        | HDD18.3     | 3064        | 522   | 435   | 377   | 251   | 129   | 66    | 34    | 37    | 99    | 223   | 384   | 506   | (9)  |
|                                                                        | CDD10.0     | 1203        | 1     | 1     | 12    | 48    | 144   | 216   | 277   | 268   | 164   | 68    | 4     | 1     | (10) |
|                                                                        | CDD18.3     | 152         | 0     | 0     | 0     | 1     | 9     | 31    | 53    | 47    | 11    | 0     | 0     | 0     | (11) |
|                                                                        | CDH23.3     | 1307        | 0     | 0     | 0     | 13    | 97    | 262   | 455   | 412   | 66    | 2     | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 353         | 0     | 0     | 0     | 0     | 17    | 56    | 132   | 145   | 3     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 2.0         | 2.3   | 2.4   | 2.5   | 2.1   | 1.9   | 1.8   | 1.9   | 1.7   | 1.6   | 1.7   | 1.8   | 2.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 1471        | 140   | 124   | 114   | 96    | 121   | 107   | 109   | 114   | 108   | 133   | 143   | 163   | (15) |
|                                                                        | PrecMax     | 1920        | 307   | 273   | 377   | 194   | 208   | 224   | 230   | 276   | 220   | 246   | 344   | 342   | (16) |
|                                                                        | PrecMin     | 1062        | 30    | 21    | 23    | 25    | 43    | 48    | 35    | 26    | 33    | 15    | 27    | 11    | (17) |
|                                                                        | PrecStd     | 221         | 73    | 68    | 82    | 51    | 44    | 52    | 45    | 56    | 50    | 61    | 76    | 82    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 11.9  | 14.8  | 20.0  | 25.2  | 28.7  | 30.4  | 32.6  | 35.5  | 27.4  | 22.7  | 16.3  | 12.5  | (19) |
|                                                                        |             | MCWB        | 9.1   | 8.9   | 11.0  | 14.9  | 19.7  | 20.0  | 20.5  | 19.5  | 18.9  | 16.6  | 12.4  | 9.8   | (20) |
|                                                                        | 2%          | DB          | 10.6  | 12.6  | 17.1  | 22.1  | 25.9  | 28.4  | 29.9  | 30.4  | 25.5  | 20.0  | 13.6  | 10.5  | (21) |
|                                                                        |             | MCWB        | 8.8   | 7.9   | 10.3  | 13.6  | 17.9  | 19.6  | 19.9  | 19.3  | 18.4  | 15.3  | 10.8  | 8.8   | (22) |
|                                                                        | 5%          | DB          | 9.2   | 10.6  | 14.6  | 19.5  | 23.8  | 26.6  | 28.1  | 27.9  | 23.4  | 18.3  | 11.8  | 9.2   | (23) |
|                                                                        |             | MCWB        | 7.5   | 7.3   | 9.2   | 12.3  | 16.1  | 19.1  | 19.4  | 19.1  | 17.6  | 14.5  | 9.6   | 7.7   | (24) |
|                                                                        | 10%         | DB          | 7.5   | 8.8   | 12.3  | 17.2  | 21.6  | 24.5  | 26.1  | 25.3  | 21.4  | 16.4  | 10.3  | 7.7   | (25) |
|                                                                        |             | MCWB        | 6.3   | 6.5   | 8.5   | 11.3  | 15.1  | 18.0  | 19.0  | 18.5  | 16.5  | 13.5  | 8.7   | 6.9   | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.6  | 9.6   | 12.5  | 15.6  | 20.3  | 21.7  | 21.9  | 21.5  | 20.5  | 17.5  | 12.7  | 10.7  | (27) |
|                                                                        |             | MCDB        | 11.1  | 13.3  | 17.4  | 23.0  | 27.7  | 27.9  | 28.9  | 30.3  | 25.1  | 21.1  | 15.3  | 11.6  | (28) |
|                                                                        | 2%          | WB          | 9.0   | 8.6   | 11.1  | 14.1  | 18.7  | 20.7  | 20.9  | 20.5  | 19.3  | 16.0  | 11.2  | 9.1   | (29) |
|                                                                        |             | MCDB        | 10.5  | 11.4  | 14.7  | 20.9  | 24.8  | 26.8  | 28.1  | 28.3  | 24.0  | 18.8  | 13.0  | 10.3  | (30) |
|                                                                        | 5%          | WB          | 7.7   | 7.8   | 10.0  | 12.9  | 17.3  | 19.7  | 20.1  | 19.5  | 18.1  | 14.8  | 10.0  | 7.9   | (31) |
|                                                                        |             | MCDB        | 8.9   | 10.3  | 13.1  | 18.4  | 22.2  | 25.4  | 27.0  | 26.3  | 22.5  | 17.6  | 11.5  | 9.0   | (32) |
|                                                                        | 10%         | WB          | 6.4   | 6.6   | 9.1   | 11.8  | 15.9  | 18.6  | 19.2  | 18.7  | 16.9  | 13.7  | 9.0   | 6.8   | (33) |
|                                                                        |             | MCDB        | 7.5   | 8.7   | 12.3  | 16.4  | 20.0  | 23.4  | 24.9  | 24.4  | 20.5  | 16.2  | 10.2  | 7.7   | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.4   | 6.5   | 8.1   | 9.7   | 9.8   | 10.3  | 10.1  | 9.8   | 9.2   | 7.8   | 5.8   | 4.6   | (35) |
|                                                                        |             | MCDBR       | 6.2   | 9.3   | 12.0  | 14.0  | 13.5  | 13.1  | 13.4  | 14.1  | 12.0  | 10.6  | 8.0   | 6.2   | (36) |
|                                                                        | 5% WB       | MCWBR       | 4.7   | 5.5   | 6.5   | 7.0   | 6.6   | 6.2   | 5.8   | 5.8   | 6.2   | 6.1   | 5.4   | 4.6   | (37) |
|                                                                        |             | MCWBR       | 5.4   | 7.9   | 9.7   | 12.9  | 12.0  | 12.2  | 12.3  | 12.2  | 10.7  | 9.7   | 7.6   | 5.6   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.306 | 0.330 | 0.361 | 0.397 | 0.404 | 0.409 | 0.407 | 0.401 | 0.378 | 0.359 | 0.325 | 0.303 | (40) |
|                                                                        |             | taud        | 2.448 | 2.383 | 2.327 | 2.243 | 2.266 | 2.288 | 2.290 | 2.321 | 2.365 | 2.405 | 2.450 | 2.450 | (41) |
|                                                                        | Ebn at Noon | Ebn at Noon | 777   | 829   | 854   | 854   | 860   | 856   | 852   | 841   | 826   | 782   | 742   | 733   | (42) |
|                                                                        |             | Edh at Noon | 66    | 87    | 107   | 128   | 130   | 128   | 126   | 117   | 102   | 84    | 66    | 59    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.05  | 1.86  | 3.09  | 4.33  | 5.05  | 5.95  | 5.66  | 4.92  | 3.72  | 2.31  | 1.21  | 0.85  | (44) |
|                                                                        | RadStd      | RadStd      | 0.14  | 0.30  | 0.48  | 0.65  | 0.51  | 0.54  | 0.51  | 0.47  | 0.36  | 0.21  | 0.14  | 0.17  | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (96 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|      |                         |       |         |        |         | +0.38 | +0.29 |             |         |         |         |

Nomenclature: See separate page